LESSON PLAN

A/C ENGINEERING SYSTEM (17)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	04
Total No. of Lesson Plans	30	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Brief description and purpose of inverted flight valve and recuperate
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Brief description and purpose of inverted flight valve and recuperate.
5. Specific Objectives	17.1.1.1 Trainee will be able to understand the construction and requirements of typical aircraft fuel system and its components.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of fuel system schematic diagrams; Instructor supervision during Q&A; session
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the aircraft fuel system?
(b) Name three components found in a typical aircraft fuel system.
(c) Why is fuel management critical during inverted flight?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of inverted flight valve • Location and construction details • Operational mechanism during inversion • Prevention of fuel flow to engine • Safety requirements for installation	L & Ds	PPT, IB	15 min
(b)	• Definition of recuperator valve • Function in fuel return system • Construction and internal components • Role in maintaining fuel pressure • Impact on engine performance	L & Ds	PPT, PDF	20 min
(c)	• Comparison of inverted valve and recuperator • System integration and interaction • Maintenance and inspection criteria • Troubleshooting common failures • Regulatory compliance and standards	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the inverted flight valve prevents fuel starvation during inverted maneuvers.
(b) Describe the role of the recuperator in the fuel return line and its effect on system pressure.

12. Home Assignment and Follow Up	Prepare a brief note on how the principles of inverted flight valves and recuperators apply to the pressure refueling and de-fueling procedures covered in LP 05.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Principles of pressure refueling and de-fueling	Refer to the master syllabus reference list	LP 05

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min